

Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring

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Abstract

Purpose: Computer-vision systems for handwriting commonly assume mutually exclusive semantic regions, optimize recognition or style classification, or compare images without separating global placement from local structural error. None returns what assessing a handwritten Chinese character against a same-character reference requires: two instances can be equally recognizable yet differ in the local stroke structure a learner should revise. We formulate reference-conditioned structural assessment as an auditable document-analysis problem under limited labelled data.

Methods: An overlapping six-channel parser predicts five independently activated direction masks and an endpoint mask; U-Net, DeepLabV3+, and SegFormer-B2 are compared with three seeds on standard and character-disjoint splits of 769 quality-controlled samples. Constrained similarity registration, three structural scores, and localized diagnostic evidence are evaluated on 200 references under controlled perturbations and on 150 natural pairs rated blindly by three calligraphy-domain raters, two of them authors.

Results: DeepLabV3+ reaches 0.9630 ± 0.0006 standard-split Macro Dice, SegFormer-B2 leads character-disjoint direction parsing at 0.8866 ± 0.0008 , and U-Net leads strict endpoint F1. Registration reduces nuisance penalties by 30.118 points relative to no alignment on clipping-safe cases while preserving 11.189 additional points of structural penalty. The frozen production score, a coverage-aware score, and a retrospectively developed aligned spatial-distribution similarity (ASDS) reach Spearman correlations of 0.297, 0.429, and 0.556; character-grouped out-of-fold ASDS retains 0.540, and direct-ink ASDS is nearly equivalent ($\Delta\rho = 0.0004$), so direction-level evidence, not the scalar, justifies the parser. Localized evidence reaches Recall@3 of 0.701 but exact-cell localization of 0.109.

Conclusion: We deliver a reproducible framework for overlapping parsing, discrepancy-preserving registration, and human-audited localized feedback.

Keywords: Chinese calligraphy, document analysis, multi-label segmentation, reference-based assessment, human validation, explainable feedback

1 Introduction

Chinese calligraphy (*shufa*) is both a writing practice and an important form of Chinese traditional

culture. Two handwritten instances of the same character are often equally legible, yet a calligraphy teacher would mark different things on each: a horizontal stroke that stops short, a hook

that departs from the wrong point, a component placed too low, an endpoint that fails to meet its neighbour. Figure 1(a) shows such a pair, a regular-script reference and a running-script candidate of one character: both are recognizable, yet they differ in stroke proportion, in where strokes meet or cross, and in the space between components. A learner benefits from knowing which of these differences matter and where they are.

Not every difference is an error. The character may be written slightly larger, shifted on the canvas, or tilted by a degree or two without any change of structure; a missing terminal fragment, an over-wide stroke, or a misproportioned component is exactly what assessment should retain. A reference-conditioned assessment therefore has to compensate global placement while preserving local discrepancy, and to expose that discrepancy as inspectable evidence rather than as a number alone.

Recognition, style classification, and generation reduce a character image to an identity, a category, or a plausible glyph [14, 25, 26]; none of them returns the spatially resolved, direction-specific comparison that this question requires. Treated as a document-analysis problem, same-character structural comparison exposes four computer-vision challenges.

- C1 *Overlapping labels under scarce annotation.* At crossings and adhesions one pixel belongs to more than one stroke, so a mutually exclusive partition cannot represent the label geometry; endpoint evidence is sparse; and dense labels are scarce (769 quality-controlled observations of 40 character identities, with no trustworthy writer identity).
- C2 *Global nuisance versus local structure.* Direct overlap confounds structure with translation, scale, and rotation; flexible registration can erase the proportion or placement error that assessment should retain; and an overlap score can reward direction channels that are empty in both images.
- C3 *Stability is not validity.* A score that is stable under allowed transforms may still disagree with how trained readers perceive structural similarity.
- C4 *A scalar does not localize.* Feedback must name the direction family, the region, and whether

ink is missing or extra; fluent text cannot repair a wrong attribution.

Figure 1 outlines the formulation studied here: the learner image and the reference are parsed into overlapping structural channels, the reference is registered to the learner with a bounded global transform, and the aligned pair yields a scalar agreement together with directional and spatial evidence that precedes any text realization. Parsing addresses C1 and registration C2; the score and the evidence outputs are the objects of C3 and C4, tested by the human study and the diagnostic benchmark. The paper asks four corresponding research questions.

RQ1 (representation and generalization). Can convolutional and transformer parsers trained on the quality-controlled corpus of Sect. 3 recover overlapping direction masks and sparse endpoint masks, and how much does performance drop for character identities never seen in training?

RQ2 (registration and score behaviour). Relative to no alignment, does a constrained similarity registration reduce sensitivity to nuisance transforms while preserving penalties for controlled structural changes; does a wider search improve that trade-off; and how do the frozen score’s channel-coverage semantics and same-character discrimination behave on real references?

RQ3 (human validity). How strongly do three fixed scores of the aligned pair, the frozen production score, the coverage-aware score, and aligned spatial-distribution similarity (ASDS), associate with blinded human ratings of same-character structural similarity, and does the ASDS scalar depend on six-channel parsing?

RQ4 (localized diagnosis). Which localized diagnostic outputs are reliable on interventions with known ground truth, and why do they fail?

The contributions follow the research questions.

1. For RQ1, a six-channel representation with five independently activated direction-family channels and an endpoint channel, needing no per-character prior at inference (unlike Stroke-Seg’s stroke instances, Sect. 2), a semantic quality-control protocol, and a matched three-seed comparison of U-Net, DeepLabV3+, and SegFormer-B2 on standard and character-disjoint splits. Ranking depends on target and

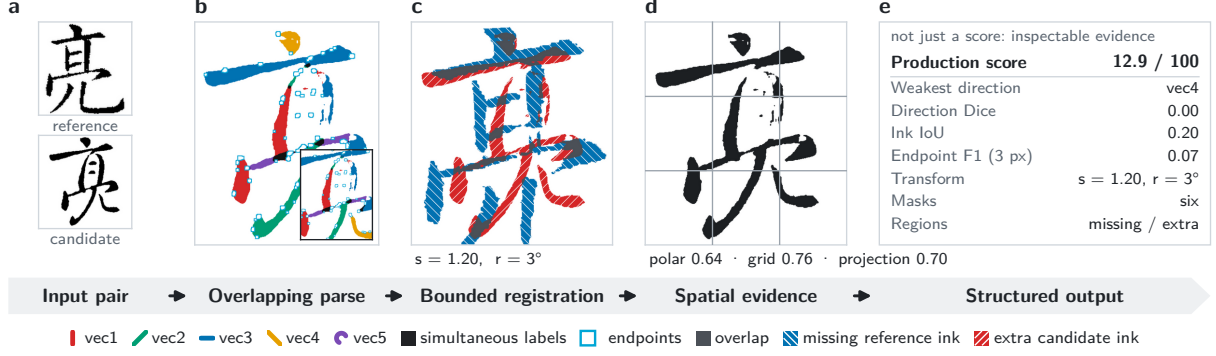


Fig. 1 Reference-conditioned structural assessment on one same-character pair. (a) A regular-script reference and a running-script candidate of the same character are recognizable but structurally different. (b) The candidate parsed into overlapping direction channels vec1–vec5 (the reference is parsed identically); black pixels carry more than one direction label, cyan outlines mark endpoints, and the framed inset shows the bounding box of the multi-label pixels at reduced scale. (c) The reference is registered to the candidate with a bounded global transform (here isotropic scale 1.20 and rotation 3°), which compensates global size and tilt; the residual is shown as overlap (dark grey), reference ink missing from the candidate (blue, hatched), and extra candidate ink (red, counter-hatched), and is the discrepancy that assessment preserves. (d) The aligned silhouette on the 3×3 grid used for localized evidence; the values beneath it are the three ASDS spatial-distribution similarities of Sect. 4.4. (e) The returned contract: production score, weakest direction family with its own Dice (not the five-channel mean), ink IoU, endpoint F1, transform, and missing/extra regions, all available before any text is generated. The cross-style pair is for illustration; the human study uses natural same-character, different-instance pairs from the project corpus. Source images: Calli-Tongji open subset [19], CC BY-NC 4.0.

protocol: DeepLabV3+ leads in-corpus direction parsing, SegFormer-B2 unseen-character direction parsing, and U-Net strict endpoint localization.

- For RQ2, a constrained similarity registration evaluated with controlled nuisance and structural perturbations of 200 real references, paired comparisons against no alignment and a wider search, and an inactive-channel audit of the score. Constrained registration substantially reduces nuisance sensitivity relative to no alignment while preserving structural penalties, with residual sensitivity and clipping-conditional summaries reported explicitly.
- For RQ3, a blinded three-rater association study of the frozen production score, the coverage-aware score, and ASDS, reported as retrospective development with character-grouped internal validation, plus a paired direct-ink control. ASDS associates most strongly with the ratings but is a silhouette measure that does not justify the six-channel parser, whose value lies in direction-level evidence.
- For RQ4, localized structured evidence with a diagnostic failure taxonomy. The required finding is usually surfaced, but exact-cell localization is unreliable, mainly because multi-cell

truth is scored by a single-cell metric and direction attribution errs; language generation stays downstream of the evidence.

The intended input is a raster exported from a direct digital writing canvas, not a photograph. The claims concern reference-conditioned structural analysis in this modality. They do not extend to camera robustness, writer-disjoint generalization, stroke-order recovery, historical authenticity, or universal aesthetic grading.

2 Related Work

2.1 Recognition, document segmentation, and stroke parsing

Large databases such as CASIA-HWDB established benchmarks for online and offline handwritten Chinese character recognition [14]. Recognition maps an image to a character identity and returns no spatial account of the strokes that compose it. Dense prediction supplies such an account. U-Net introduced an encoder–decoder with skip connections [18]; DeepLabV3+ combines atrous spatial pyramids with a decoder [3]; and SegFormer pairs hierarchical transformer features with a lightweight multilayer-perceptron

decoder [24]. In document analysis, dhSegment showed that one generic segmentation formulation can serve many page-level tasks [17], and a recent IJDAR study found that label scarcity, source–target mismatch, and the pretraining protocol materially affect handwritten document segmentation [1]. Standard semantic segmentation assigns each pixel one class [3, 24].

Stroke-level analysis of Chinese characters has been addressed by Stroke-Seg, a framework compatible with several segmentation backbones that adds a multi-label output so that pixels in intersection regions can be assigned to more than one stroke and supplies character-specific stroke-composition information to the network as a prior vector; it is evaluated by the true stroke rate on a brush-calligraphy stroke-segmentation dataset [9].

2.2 Computational calligraphy and handwriting-quality evaluation

Early computational calligraphy work represented style with local and global shape features [8, 27]; CalliNet advances style classification with metric learning and triplet supervision [26]. Other systems extract artistic regions from a calligraphy work relative to a reference image [20], beautify handwritten glyphs toward a Kai-style template [23], or evaluate robotic writing with designed aesthetic features [15]. Large recent resources such as UniCalli support column-level generation and recognition [25].

Quality evaluation of handwritten Chinese characters has been surveyed as three families, rule-based geometric measures, template comparison, and learned methods, with the difficulty of defining quality labels and explaining model decisions identified as open problems [4]. Siamese networks learn pairwise quality from labelled examples [5], and a recent shared task combined writing-quality grading with comment generation on an expert-annotated dataset [21]. Across these studies, quality is not a single observable: legibility, geometric correctness, similarity to a reference, brushwork, and artistic appeal can disagree.

2.3 Registration, shape description, and human validation

Image registration ranges from parametric geometric alignment to learned spatial transformers

[10, 28]. Registration surveys distinguish rigid, similarity, and affine models from elastic and deformable models by how much shape difference they absorb [28]. Shape contexts describe the spatial distribution of contour points in log-polar bins [2], and spatial pyramids retain coarse layout that an orderless representation would discard [12]. Distributions of this kind can be compared with the bounded, symmetric Jensen–Shannon divergence [13].

Automated assessment meant to stand in for human judgment requires both an association with that judgment and an account of rater reliability, for which two-way random-effects, absolute-agreement intraclass correlations are the established statistics [11, 16]; bootstrap resampling provides uncertainty for correlation estimates [7].

3 Data Resources

The experiments draw on three sources: a project corpus with manual ground truth (GT) after quality control (QC), an external reference library whose model-parsed masks feed three derived cohorts, and a human-rating cohort. Table 1 lists the resulting resources and the research question each serves; the subsections define them.

3.1 Project-collected corpus and six-channel annotation contract

The segmentation corpus was collected in an earlier phase of the project and recovered from its acquisition archive for this study. It was commissioned from contributors described as calligraphy-trained, although no record of individual training survives, who wrote complete characters and, separately, each of their component strokes on a direct digital canvas, marking the two endpoints of every isolated stroke. The six-channel target is composed from these captured components (Fig. 2): a fixed, character-specific mapping assigns each isolated stroke to one of five direction families, and the pixels of strokes in the same family are merged into one mask. Because the strokes were acquired independently, a crossing or adhesion can occupy the same pixel in two families, so the composed masks overlap and a crossing pixel legitimately carries more than one direction label. The sixth channel is the union of the manually marked endpoints of all isolated strokes. It encodes

Table 1 Data resources, their label status, and the research question (RQ) each serves. GT denotes manual six-channel ground truth.

Resource	Origin	Size	Label status	RQ
Project corpus	commissioned direct-digital acquisition	769 QC-clean of 840 complete; 40 identities	GT	1, 3
Standard split	frozen sample assignment	530/119/120 train/val/test	GT	1
Character-disjoint split	frozen identity assignment, 28/6/6	539/114/116 train/val/test	GT	1
Reference library	Calli-Tongji Beta, two style categories	200 images	model-derived cache, not GT	2, 4
Perturbation / diagnostic cohort	deterministic edits of cached masks	200 references $\times$ 11 perturbations (5 nuisance, 6 structural)	known intervention	2, 4
Cross-reference cohort	reference library	7 positive, 50 negative pairs	pair type only	2
Human-rating cohort	natural pairs from the project corpus	150 pairs + 15 repeats, 3 raters	blinded ratings; development data	3

stroke termini only, not junctions, turning points, stroke order, or generic salient points. No label in the corpus is produced by any model evaluated in this paper.

The archive records preserve neither a trustworthy mapping from samples to writers nor a standardized measure of contributor expertise, so the paper makes no professional-calligrapher, different-writer, or writer-disjoint claim.

3.2 Quality control and split protocols

Recovery of the acquisition archive yielded 894 sample directories across 43 character identities. Complete six-channel ground truth is available for 840 samples from identities 0–39; the 54 samples of identities 40–42 lack the character-specific direction mapping and are excluded rather than pseudo-labelled. A frozen semantic quality-control pass then identified 12 source-image/ground-truth mismatches and 59 non-canonical members of exact image-and-mask duplicate groups. The two exclusion sets do not overlap, leaving 769 unique QC-clean observations of 40 identities (Fig. 3e,f). All exclusion decisions were fixed before the formal model comparison.

Two split protocols are used. The *standard* assignment was frozen before quality control by per-character sample index, with the same index ranges in every identity, and contains 600/120/120 train/validation/test samples before quality control and 530/119/120 after it; it measures in-corpus generalization to new instances of seen identities. The *character-disjoint* assignment was

frozen at the identity level before any formal evaluation, with 28/6/6 disjoint identities and 588/126/126 samples before quality control and 539/114/116 after it; it measures transfer to identities never seen in training. No exact-duplicate group crosses either split.

3.3 External reference library and derived cohorts

The reference-conditioned experiments use 200 approved images from the publicly downloadable open subset of Calli-Tongji [19] (50 calligrapher-script categories with 100 samples each, released under CC BY-NC 4.0): the 100 images of the Ouyang Xun regular-script category and the 100 images of the Wang Xizhi running-script category. Each reference is parsed once by a frozen SegFormer-B2 checkpoint (version v1, trained with the recipe of Sect. 5.1 on the pre-quality-control standard assignment and frozen before the formal three-seed comparison), and the resulting six binary masks are cached with their provenance. These cached masks are model-derived reference representations; they are never treated as segmentation ground truth and never enter parser training.

Two cohorts are derived deterministically from the cached masks. The *perturbation cohort* applies the nuisance and structural edits of Sect. 5.2 to every reference, so that registration and score behaviour can be studied without a second parser inference. The *cross-reference cohort* contains all seven natural same-character cross-style pairs that

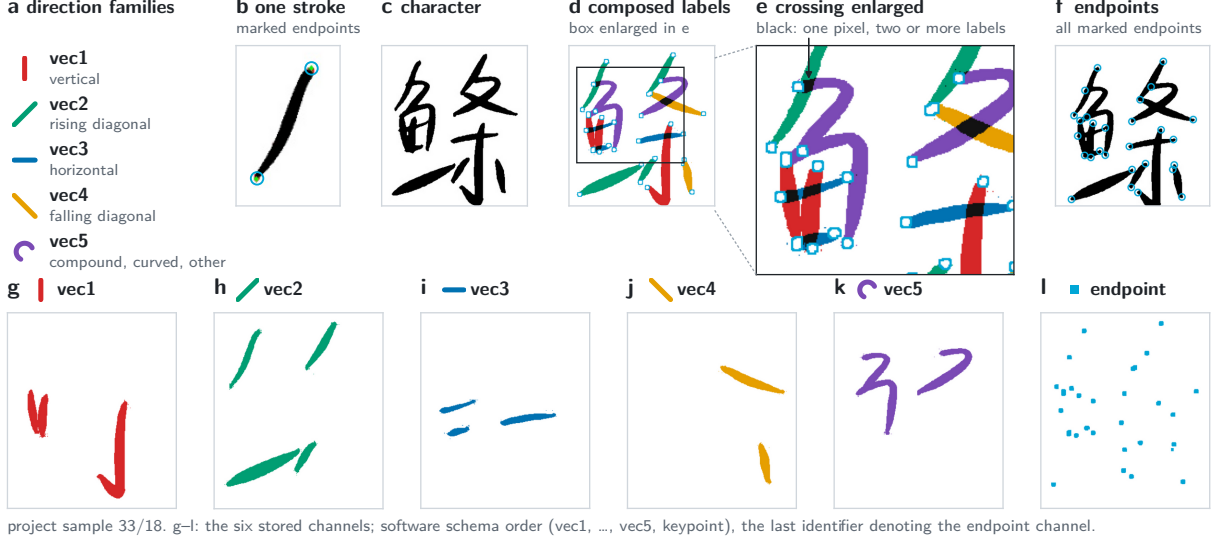


Fig. 2 Six-channel overlapping annotation contract on one project sample. (a) Direction families: vec1 vertical, vec2 rising diagonal, vec3 horizontal, vec4 falling diagonal, vec5 compound, curved, or otherwise non-cardinal strokes. (b) An isolated stroke source with its two marked endpoints (circles are display only); (c) the complete character; (d) the composed annotation, where black pixels carry simultaneous direction labels; (e) the boxed region of (d) enlarged, the arrow marking a pixel that carries two direction labels at once; (f) the endpoint channel, the union of the marked stroke endpoints (cyan outlines and dilation are display only); (g)–(l) the six channels. The software schema stores the channels as (vec1, ..., vec5, keypoint); the last identifier is retained for backward compatibility and denotes this endpoint channel.

the library supports (Fig. 3g) and 50 different-character negatives selected independently of any model score; the library contains no same-style, different-instance reference pair. The perturbation cohort also serves as the diagnostic cohort of Sect. 5.4, where each known intervention is the ground truth.

3.4 Human-rating development cohort

Before any rating was collected, 150 natural same-character, different-instance pairs were selected from the 769 QC-clean project samples, three or four per identity, excluding mismatches, exact and suspected near duplicates, and repeated instances. Pair roles, asset hashes, blinded identifiers, and display order were frozen, and all automated scores and source identifiers were hidden from the raters.

Three adult raters independently judged structural similarity on a five-point ordinal scale, each rating the 150 canonical pairs plus 15 interspersed hidden repeats. The panel comprised two author-raters with prior calligraphy study, one of whom holds a Level-9 calligraphy certificate, and one non-author rater with master’s-level

training in Fine Arts (Calligraphy). The paper describes them as domain-qualified, calligraphy-domain raters; the panel is small and partly author-involved, so it is not an independent expert cohort. All scores on this cohort are computed from the annotated six-channel masks of both instances, with no parser in the loop, so they measure score validity without parser error. The ratings are the ASDS development data of Sect. 5.3, not an external test set.

4 Method

4.1 Overview and relation to prior formulations

Let I_u be a learner image and I_r an approved reference image of the same target character, one that passed the licence and quality review of the reference library (Sect. 3.3); in deployment the system never scores images of different characters. The system parses both images into overlapping structural masks, registers the reference to the learner with a restricted global transform, and returns (i) six masks per image, (ii) a scalar structural-agreement score, and (iii) localized evidence (Fig. 1). The learner image is called

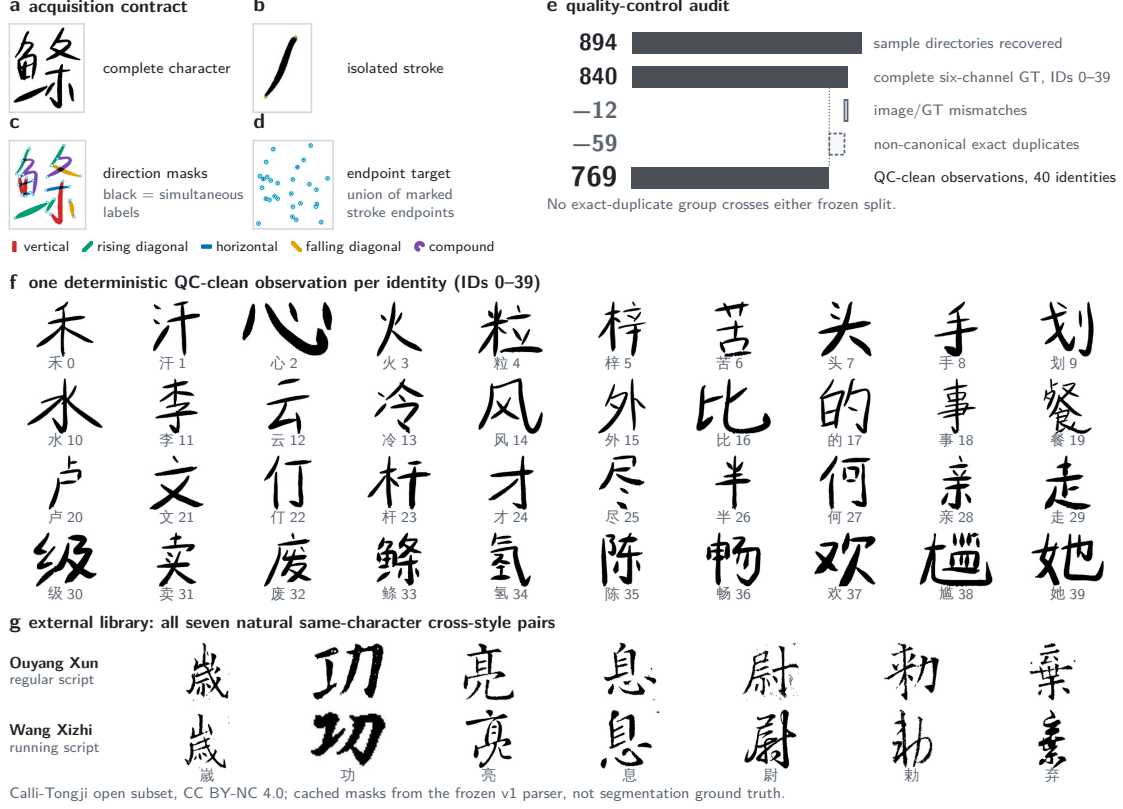


Fig. 3 Corpus, quality control, and external reference library. (a)–(d) The acquisition contract (as in Fig. 2); (e) the quality-control audit (bar length proportional to sample count: 894 recovered, 840 with complete six-channel ground truth for identities 0–39, 769 QC-clean after removing 12 image/GT mismatches and 59 non-canonical exact duplicates), with no exact-duplicate group crossing either frozen split; (f) one deterministic QC-clean observation per identity (IDs 0–39); (g) all seven natural same-character cross-style pairs in the external library, represented by cached masks from the frozen v1 parser, not by segmentation ground truth. Source images: Calli-Tongji open subset [19], CC BY-NC 4.0.

the candidate in the figures and `user` in the software contract; the terms denote the same image. Three decisions distinguish this formulation from the work of Sect. 2. Unlike Stroke-Seg [9], which recovers individual strokes with a character-specific composition prior supplied as input, the parser predicts five direction families that may overlap plus a separate endpoint channel, uses no per-character prior at inference, and feeds a reference-conditioned comparison rather than a stroke inventory. The comparison is not a learned pairwise similarity network [5]; every score is a fixed, decomposable function of aligned masks whose semantics can be audited. Registration is a bounded similarity transform rather than a learned or deformable warp [10, 28] (Sect. 4.3).

4.2 Overlapping six-channel parser

For an RGB character image I , a parser f_θ predicts

$$P = f_\theta(I) \in [0, 1]^{H \times W \times 6},$$

whose channels follow the annotation contract of Sect. 3.1: five direction families and the endpoint channel. Because a crossing pixel may carry several direction labels, each channel is activated by an independent sigmoid rather than by a mutually exclusive softmax; the five direction outputs may therefore overlap, exactly as the composed ground truth does. For direction channel c , the loss combines class-weighted binary cross entropy (BCE)

and soft Dice, averaged over the five channels,

$$\mathcal{L}_{\text{dir}} = \frac{1}{5} \sum_{c=1}^5 \left(\mathcal{L}_{\text{BCE}}^{(c)} + \mathcal{L}_{\text{Dice}}^{(c)} \right).$$

The sparse endpoint target uses focal loss with $\gamma = 2$ and $\alpha = 0.75$ plus Dice loss, and a boundary term derived from the masks is weighted by 0.2:

$$\mathcal{L} = \mathcal{L}_{\text{dir}} + \mathcal{L}_{\text{focal, end}} + \mathcal{L}_{\text{Dice, end}} + 0.2 \mathcal{L}_{\text{boundary}}.$$

Binary masks are obtained from P with per-channel thresholds calibrated on validation data only (Sect. 5.1).

4.3 Reference cache and constrained registration

Let U and R denote the learner and cached reference binary stacks (Sect. 3.3) and let

$$M(X) = \bigvee_{c=1}^5 X_c$$

be the union of direction ink. For each candidate isotropic scale s and rotation ϕ , the transformed reference is translated so that its foreground centroid approaches the learner centroid. The production search evaluates nine evenly spaced scales and seven evenly spaced rotations in

$$s \in [0.8, 1.2], \quad \phi \in [-3^\circ, 3^\circ],$$

and selects the candidate maximizing the foreground intersection over union (IoU) $J(M(U), M(T_{s, \phi, t}(R)))$. Non-uniform scale, shear, thin-plate splines, and deformable warping are disallowed. This bounded search is a conservative compensation policy: it can absorb page placement and modest global size or rotation differences while leaving local proportion and stroke-region errors available for diagnosis; it implies no invariance within the nominal range, and residual sensitivity is measured in Sect. 6.2.

4.4 Structural agreement scores

Three scores are studied. Each is a fixed function of the aligned masks and uses no character identity, writer identity, style identity, or human rating at inference.

Frozen production score. After alignment, let D_c be Dice agreement in direction channel c , $\overline{D} = \frac{1}{5} \sum_{c=1}^5 D_c$, let J_{ink} be union-ink IoU, and let $F_{\text{end}}^{(3)}$ be endpoint-mask F1 with a three-pixel tolerance. The deployed score is

$$S_{\text{prod}} = 100 \left(0.55 \overline{D} + 0.25 J_{\text{ink}} + 0.20 F_{\text{end}}^{(3)} \right).$$

It was frozen before the controlled experiments and is never retuned in this paper.

Coverage-aware score. The production Dice convention returns 1 for a channel that is empty in both images, so a character that uses fewer direction families can receive credit for structure that is absent. The coverage-aware score keeps the same evidence families and weights but averages Dice only over direction channels active in at least one image, and treats both-empty endpoint masks as unavailable evidence with the remaining weights renormalized. It is reported alongside S_{prod} as an audit variant and does not replace it.

Aligned spatial-distribution similarity (ASDS). Pixel- and channel-overlap terms say little about how ink mass is organized across the canvas. ASDS therefore compares three spatial distributions of the union silhouette $M(\cdot)$ after the same frozen registration. *Polar occupancy* centres the foreground at its ink centroid, divides radius by the 95th percentile of foreground radius (clipped to $[0, 1]$), and assigns pixels to four radial and eight angular bins, giving a normalized 32-bin histogram h_p ; by construction, centroid and radius normalization make it insensitive to residual translation and isotropic rescaling of the aligned silhouette, a property not separately measured here. *Grid occupancy* h_g counts normalized foreground over a 3×3 partition of the canvas and retains coarse location and balance. *Projection profiles* h_x and h_y are the normalized column and row sums. For distributions p and q with $m = (p + q)/2$, the Jensen–Shannon divergence and the Jensen–Shannon similarity (JSS) are

$$D_{\text{JS}}(p, q) = \frac{1}{2} D_{\text{KL}}(p \| m) + \frac{1}{2} D_{\text{KL}}(q \| m),$$

$$\text{JSS}(p, q) = 1 - \sqrt{D_{\text{JS}}(p, q)},$$

with base-2 logarithms, so that $\text{JSS} \in [0, 1]$ [13]. With $s_p = \text{JSS}(h_p(U), h_p(R))$, $s_g =$

$JSS(h_g(U), h_g(R))$, and s_{proj} the mean of the column and row similarities, the frozen specification is

$$S_{\text{ASDS}} = 100(0.70 s_p + 0.15 s_g + 0.15 s_{\text{proj}}).$$

ASDS was developed after the ratings of Sect. 3.4 were available and its weights were selected on them (Sect. 5.3). Because it consumes only the union silhouette, it does not use the direction semantics of the six-channel masks; whether those masks contribute anything to this scalar beyond their silhouette is tested by the direct-ink control (Sect. 6.3).

4.5 Localized evidence and feedback

Alongside the scores, the system preserves the selected transform, the pre-alignment centre distance and ink-area ratio, per-direction Dice, and endpoint agreement. Difference masks between the aligned stacks are summarized by direction channel, by type (reference ink missing from the learner versus extra learner ink), and by a 3×3 region set; connected components are exposed as direction-classified regions, not as recovered strokes or stroke order. Deterministic rules rank centre, scale, endpoint, and local-direction findings. An optional text model receives only this structured contract; it may improve fluency but cannot change the score or add brushwork or stroke-order claims the evidence does not contain.

5 Experimental Setup

This section states how each research question is tested; the cohorts themselves are defined in Sect. 3.

5.1 Parser training and evaluation (RQ1)

Three architectures are compared: a U-Net with 32 base channels, a ResNet-50 DeepLabV3+, and SegFormer-B2. U-Net has no external pre-training; all of its parameters are randomly initialized. DeepLabV3+ loads the torchvision default ImageNet-1K weights for its ResNet-50 encoder, while its atrous spatial pyramid pooling (ASPP) module, low-level projection, decoder, and six-channel output head

are randomly initialized. SegFormer-B2 is initialized from `nvidia/segformer-b2-finetuned-ade-512-512`: the compatible Mix Transformer (MiT-B2) encoder and decode-head parameters are loaded from the ADE20K-fine-tuned checkpoint, the incompatible 150-class classifier is replaced by a randomly initialized six-channel classifier, and all layers are then fine-tuned. Each architecture is trained with seeds 20260811, 314159, and 271828 on both split protocols, giving 18 formal runs.

Inputs are letterboxed to 512×512 and normalized, and masks are resized by nearest-neighbour interpolation. Batch size is 8 for U-Net and 4 for the two larger models. Label-safe augmentation comprises translation up to 12 pixels, isotropic scale 0.9–1.1, brightness 0.85–1.15, contrast 0.9–1.1, and Gaussian blur with probability 0.15. All models use AdamW with weight decay 0.01, automatic mixed precision, gradient clipping at 1.0, three warm-up epochs, cosine decay to 10^{-6} , at most 120 epochs, and early stopping with patience 15. U-Net uses learning rate 3×10^{-4} ; DeepLabV3+ uses encoder/decoder rates $10^{-4}/10^{-3}$; SegFormer-B2 uses $3 \times 10^{-5}/3 \times 10^{-4}$. Positive BCE weights are estimated from at most 200 training samples and capped at 100. Per-channel thresholds are calibrated on the validation split only, and the test split is evaluated once per run.

Direction quality is reported as five-direction Macro Dice and Macro IoU between thresholded predictions and ground truth, and as Boundary F1, the F1 between the one-pixel morphological boundaries of prediction and ground truth averaged over the five direction channels; endpoint quality is reported as strict endpoint F1, the pixel-exact F1 on the endpoint channel, unlike the three-pixel-tolerant term inside the production score. Test results are summarized as the mean and sample standard deviation over the three seeds. The standard protocol measures in-corpus instance generalization; the character-disjoint protocol measures transfer to unseen characters.

5.2 Registration and score-behaviour experiments (RQ2)

Controlled perturbations are applied directly to the 200 cached reference masks. The nuisance family, for which a useful registration should show little sensitivity, comprises translation, rotation, scale up, scale down, and a compound transform. The structural family, for which the score should decrease with severity, comprises terminal deletion, added direction fragment, local fragment shift, direction dilation, direction erosion, and endpoint shift. Each perturbation is applied at four severity levels, except rotation and the compound transform, which use three, so a perturbation with no invalid case contributes 800 rows from the 200 references. The affected channel and axis are selected by stable hashes, never by model error or manual choice. A transformed case that risks canvas clipping is marked invalid with a recorded reason and excluded from valid-score summaries; its fraction is reported. For every perturbation we report valid N , mean drop with a 95% reference-level bootstrap confidence interval (CI), invalid fraction, and adjacent-severity monotonicity, the fraction of adjacent severity steps at which the score does not increase; median, standard deviation, percentiles, and severity correlations are given in Online Resource 1, Table S1.

The identical reference/perturbation/severity tuples are then scored under three alignment policies: no alignment, the production constrained search of Sect. 4.3, and a wider similarity search over the same grid with $s \in [0.6, 1.4]$ and $\phi \in [-12^\circ, 12^\circ]$, fixed in the frozen experiment contract before any result was computed. Comparisons are paired by reference. Confidence intervals resample references; Wilcoxon signed-rank tests [22] and rank-biserial correlation (RBC) effect sizes use per-reference mean paired differences. Family-level comparisons are the primary ablation; perturbation-level results are descriptive and unadjusted (Online Resource 1, Table S2).

Two semantic audits accompany the ablation. The inactive-channel audit counts references with empty direction channels and recomputes the coverage-aware score of Sect. 4.4 on every valid perturbation row (Online Resource 1, Table S3).

The cross-reference test scores the seven same-character cross-style pairs and the 50 different-character negatives with S_{prod} and summarizes their separation with Cliff’s δ [6]; self-pairs are prohibited.

5.3 Human-association study and ASDS development (RQ3)

Each score is compared with the three-rater mean on the 150 canonical pairs by Spearman correlation, with 95% intervals from 10,000 bootstrap resamples clustered by target character [7]. Two-sided p values for $\rho = 0$ are reported for the two frozen scores. Inter-rater reliability uses two-way random-effects, absolute-agreement intraclass correlation coefficients ICC(2,1) and ICC(2,k) [11, 16]. Hidden repeats are excluded from correlation and ICC and are used only for exact agreement, agreement within one point, mean absolute difference (MAD), and quadratic weighted kappa (QWK).

Post-rating ASDS development. The original ratings are the ASDS development set. Three fixed distribution families (polar, grid, projection) are considered, and a coarse non-negative simplex search evaluates component weights against the ratings. The search-selected polar weight is retained, and the remaining weight is split equally between grid and projection to reduce sample-specific tuning; the full candidate search is reported in Online Resource 1, Note S5. Five-fold internal validation groups pairs by character identity and selects the weights of each fold on the other characters only, giving a character-grouped out-of-fold estimate. Character-cluster bootstrap intervals are reported for the frozen score, the out-of-fold predictions, the components, and paired differences from the production and coverage-aware scores; these comparisons are exploratory and not multiplicity-adjusted.

Paired direct-ink control. To test whether the ASDS scalar itself requires six-channel masks, ASDS is recomputed on the same 150 pairs with foreground taken directly from the original rasters instead of from the union of the annotated direction masks: grayscale pixels with intensity below 240, the threshold already frozen for dataset quality control, define ink, with no learned pre-processing. Direct-ink masks pass through the

same constrained registration and frozen formula; nothing is selected against the ratings for this comparison. We report the Spearman association of the annotated-mask-union variant (the “parsed union” of the frozen artifact) and the direct-ink variant with the ratings, their paired difference, and their mutual rank association, with 10,000 paired resamples clustered by character. Because the control reuses the development pairs, it measures implementation dependence within the same retrospective cohort, not external validity.

5.4 Diagnostic evaluation (RQ4)

The diagnostic cohort applies every severity level of the 11 perturbations to the 200 references; the known perturbation of each case supplies deterministic ground truth for the direction channel, the missing/extra type, and the affected 3×3 cells. The required finding is the centre-offset finding for translation, the ink-scale finding for scale changes, a local direction finding for the five direction edits, and the endpoint-relation finding for endpoint shifts; rotation and compound cases have no required finding. We report required Recall@3, the fraction of cases whose required finding appears among the top three returned findings, and exact grid localization, which counts a case as correct only when the returned cell equals the truth cell set and is scored on the cases whose perturbation edits a direction channel. Each exact-localization failure is assigned one primary cause in the fixed order local finding absent from the top three, wrong direction channel, wrong missing/extra type, multi-cell or grid-boundary truth, and alignment residual shift; direction and type errors are characterized only through this classification, and language-model realization is disabled.

6 Results and Discussion by Research Question

Each subsection names the experiments it draws on, reports and interprets the results, and closes with a direct answer to the research question and the boundary of the corresponding contribution. All values are computed from frozen experiment artifacts; Online Resource 1, Note S6, lists the integrity hashes of those included in the code release, and none was adjusted after inspection.

6.1 RQ1 (representation and generalization)

RQ1 is answered by the 18 formal runs of Sect. 5.1 on the standard and character-disjoint splits (Table 2) and by the qualitative comparison of Fig. 4.

On the standard split, DeepLabV3+ obtains the highest direction Macro Dice (0.9630 ± 0.0006), Macro IoU (0.9287 ± 0.0012), and Boundary F1 (0.8443 ± 0.0028). U-Net is weaker on direction and boundary metrics yet gives the best strict endpoint F1 (0.7562 ± 0.0038). Dense direction segmentation and sparse endpoint localization therefore favour different architectures.

All models degrade when character identities are held out (Table 2, bottom). SegFormer-B2 gives the strongest character-disjoint direction result: Macro Dice 0.8866 ± 0.0008 , Macro IoU 0.7977 ± 0.0013 , and Boundary F1 0.6851 ± 0.0046 . Its Macro-Dice decrease from the standard split is 7.150 percentage points, versus 9.664 for DeepLabV3+ and 9.360 for U-Net. DeepLabV3+ remains second for direction overlap, and U-Net retains the highest strict endpoint F1 (0.6972 ± 0.0045). Near-saturated in-corpus overlap thus does not imply transfer to unseen character structures, and the architecture that transfers best is not the one that fits the corpus best.

Figure 4 makes the output contract visible: at the crossing in the first row every model emits overlapping labels rather than choosing one family, and the third row shows the errors characteristic of an identity absent from training.

Answer to RQ1. Trained on the 530- and 539-sample training partitions of a 769-sample quality-controlled corpus, all three parsers reach in-corpus Macro Dice of at least 0.9196; agreement at multi-label crossing pixels is not measured separately and is shown only qualitatively in Fig. 4; strict endpoint localization is harder (best F1 0.7562). Holding out identities costs between 7.150 and 9.664 Macro-Dice points. The best model depends on the target and on the protocol: DeepLabV3+ for in-corpus direction parsing, SegFormer-B2 for unseen-character direction parsing, and U-Net for strict endpoints. No architecture is best on every metric of both split protocols. This is the first contribution; its boundary is that character-disjoint

Table 2 Segmentation results over three fixed random seeds on the QC-clean standard split and the character-disjoint split (train, validation, and test identities disjoint). Values are mean $\pm$ sample standard deviation; Endpoint F1 is the strict pixel-level score; bold marks the best value in each column within a split.

Split	Model	Macro Dice	Macro IoU	Endpoint F1	Boundary F1
Standard	U-Net	0.9196 $\pm$ 0.0018	0.8515 $\pm$ 0.0030	0.7562 $\pm$ 0.0038	0.8009 $\pm$ 0.0023
	DeepLabV3+	0.9630 $\pm$ 0.0006	0.9287 $\pm$ 0.0012	0.7218 $\pm$ 0.0012	0.8443 $\pm$ 0.0028
	SegFormer-B2	0.9581 $\pm$ 0.0031	0.9196 $\pm$ 0.0057	0.7139 $\pm$ 0.0046	0.8259 $\pm$ 0.0088
Character-disjoint	U-Net	0.8260 $\pm$ 0.0161	0.7120 $\pm$ 0.0225	0.6972 $\pm$ 0.0045	0.6430 $\pm$ 0.0136
	DeepLabV3+	0.8664 $\pm$ 0.0094	0.7663 $\pm$ 0.0146	0.6755 $\pm$ 0.0031	0.6728 $\pm$ 0.0138
	SegFormer-B2	0.8866 $\pm$ 0.0008	0.7977 $\pm$ 0.0013	0.6604 $\pm$ 0.0033	0.6851 $\pm$ 0.0046

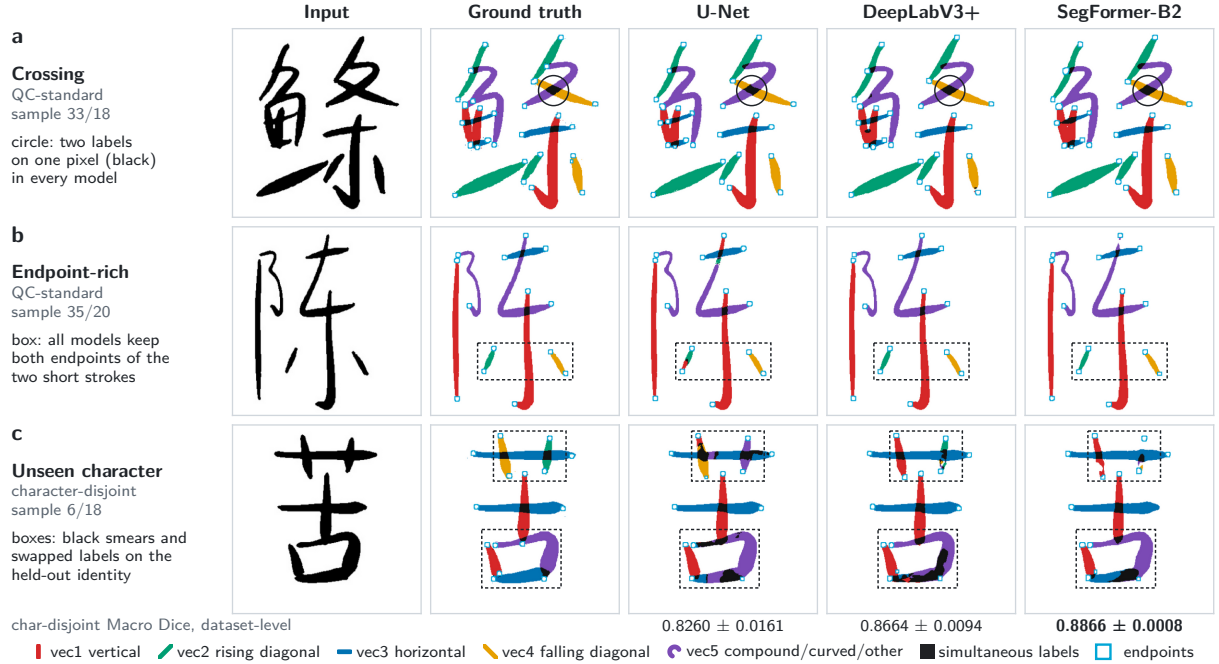


Fig. 4 Frozen qualitative comparison for seed 20260811 on a multi-channel crossing, an endpoint-rich case, and a held-out character (samples are identified as identity/sample index). The first two rows use standard-split models and the third uses character-disjoint models. Call-outs are identical across columns: the circle in (a) marks a crossing where every model emits two labels, the box in (b) two short strokes whose endpoints every model keeps, and the boxes in (c) black smears and swapped labels on the held-out identity. Values under (c) are the dataset-level character-disjoint Macro Dice of Table 2. Colours follow Fig. 2; thresholds are validation-calibrated.

evaluation measures transfer to unseen identities within one acquisition source, not to unseen writers.

6.2 RQ2 (registration and score behaviour)

RQ2 is answered by the controlled-perturbation summary (Table 3), the paired alignment ablation (Table 4, Fig. 5), the inactive-channel audit, and the cross-reference test (Online Resource 1, Table S4) described in Sect. 5.2.

Under the production registration, translation is recovered exactly in all 800 valid cases, whereas residual sensitivity remains for the other transforms: mean penalties are 5.395 points for rotation, 13.653 for scale up, and 16.356 for scale down. Scale up and the compound transform have invalid fractions of 49.0% and 54.0% because of clipping risk, so their summaries hold only for clipping-safe references and establish no invariance across the library. Among structural perturbations, direction dilation and erosion produce the largest penalties, followed by terminal deletion and added fragments; local fragment shifts

Table 3 Controlled perturbation summary. Nuisance rows report absolute score drop; structural rows report signed drop from identity; 95% CIs are reference-level bootstrap intervals; Monotonic is the fraction of non-increasing adjacent severity steps. Full distribution and severity statistics are in Online Resource 1, Table S1.

Perturbation	Valid N	Mean drop	95% CI	Invalid fraction	Monotonic
Translation	800	0.000	[0.000, 0.000]	0.000	1.000
Rotation	520	5.395	[5.079, 5.747]	0.133	0.647
Scale up	408	13.653	[12.964, 14.420]	0.490	0.271
Scale down	800	16.356	[15.670, 17.074]	0.000	0.500
Compound transform	276	8.143	[7.621, 8.717]	0.540	0.466
Terminal deletion	800	19.757	[17.387, 22.087]	0.000	0.992
Added direction fragment	800	11.398	[9.934, 12.929]	0.000	0.987
Local fragment shift	800	4.310	[3.886, 4.772]	0.000	0.985
Direction dilation	800	24.034	[22.180, 25.955]	0.000	1.000
Direction erosion	800	22.225	[20.269, 24.344]	0.000	0.997
Endpoint shift	800	7.468	[7.294, 7.641]	0.000	1.000

and endpoint shifts are weaker but monotonic in severity.

The paired comparison isolates what registration buys. Relative to no alignment, the constrained search reduces the nuisance penalty by 30.118 points (reference-bootstrap 95% CI 29.506–30.727) and simultaneously preserves 11.189 additional points of structural penalty (10.094–12.322); both per-reference signed-rank comparisons are nonzero with rank-biserial effect size 1.0. Without alignment the same structural edits cost only 3.677 points on average against 14.865 under registration (Fig. 5c), so the constrained search does not suppress structural penalties; part of the increase may reflect the centroid-derived translation re-fitting the whole reference after a local edit. The wider search incurs a 2.896-point higher nuisance penalty (2.757–3.034) while preserving only 0.284 additional points of structural penalty, so enlarging the transform range does not improve the measured trade-off (perturbation-level results in Online Resource 1, Table S2).

Forty-one of the 200 references contain inactive direction channels (36 with one and five with two; Online Resource 1, Table S3). Both-empty agreement affects 1,252 of the 7,604 valid perturbation rows, and averaging over active channels changes S_{prod} by -0.289 points on average and by at most -15.402 : macro overlap can reward absent structure, which motivates the coverage-aware score tested in RQ3.

In the cross-reference test (Online Resource 1, Table S4), the seven same-character cross-style pairs score 20.239 on average (95% CI 15.639–25.069) against 9.741 (8.584–10.970) for the 50 different-character negatives, with Cliff’s $\delta = 0.84$.

The score carries same-character signal across styles, but the small positive set and the missing same-style condition preclude a broader claim.

Answer to RQ2. Relative to no alignment, the constrained similarity registration substantially reduces nuisance sensitivity (30.118 points) while preserving structural penalties (11.189 additional points), and the wider search does not improve this trade-off. The claim is relative and bounded: rotation and scale retain residual penalties, and the scale-up and compound summaries are conditional on clipping-safe references. On real references the frozen score separates same-character from different-character pairs, but its macro overlap rewards channels that are empty in both images, so label-coverage semantics must be audited whenever the active multi-label set varies across samples. This is the second contribution.

6.3 RQ3 (human validity)

RQ3 is answered on the 150-pair human-rating cohort of Sect. 3.4 with the statistics of Sect. 5.3: association of the frozen scores (Table 5), rater reliability (Table 6), ASDS development (Table 7), and the direct-ink control (Table 8, Fig. 6).

The frozen production score correlates modestly with the three-rater mean ($\rho = 0.297$, character-cluster 95% CI 0.148–0.441), and the coverage-aware score is stronger ($\rho = 0.429$, 0.313–0.535); their paired difference is 0.132 (0.062–0.197). Single-rater reliability is below the 0.50 threshold that Koo and Li [11] label moderate (ICC(2,1) 0.448), whereas the three-rater mean is moderate (ICC(2, k) 0.709); hidden repeats yield

Table 4 Paired alignment comparison. Positive benefit differences favour the current constrained alignment and are computed from unrounded means. Ours is the production constrained search of Sect. 4.3; Comparator is no alignment (none) or the wider search (wide); values are mean score drops; RBC is the rank-biserial correlation. Wilcoxon tests use per-reference mean paired differences.

Comparison	Family	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	nuisance	200	8.455	38.573	[29.506, 30.727]	< 0.001	1.000
Ours–none	structural	200	14.865	3.677	[10.094, 12.322]	< 0.001	1.000
Ours–wide	nuisance	200	8.455	11.351	[2.757, 3.034]	< 0.001	1.000
Ours–wide	structural	200	14.865	15.149	[-0.355, -0.213]	< 0.001	-0.683

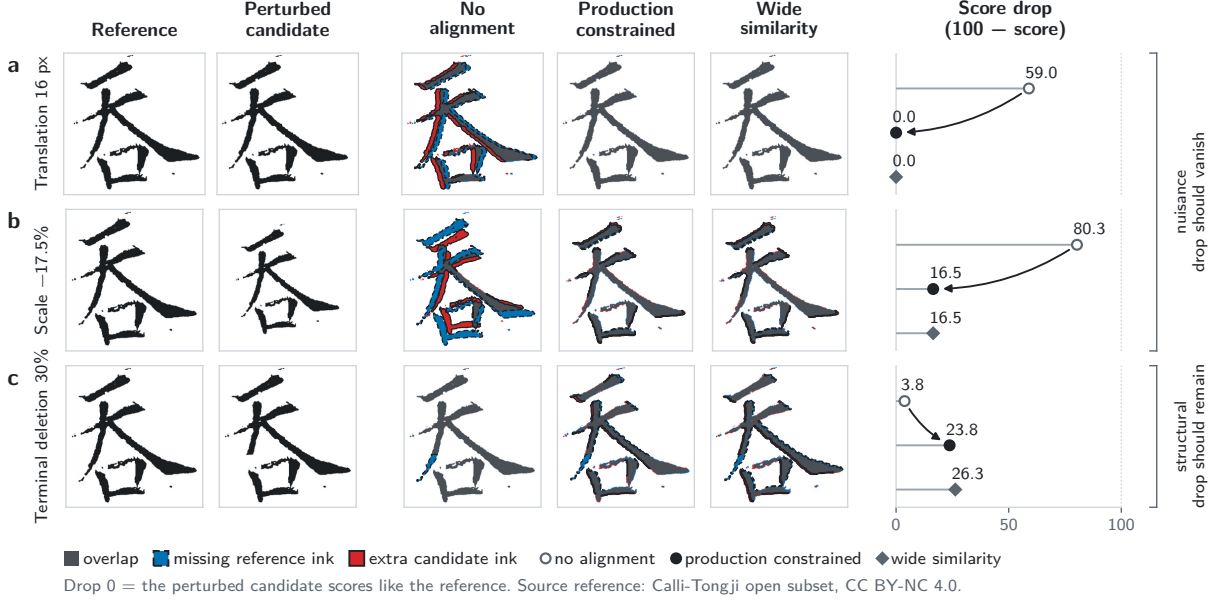


Fig. 5 Alignment ablation on deterministic translation, scale-down, and terminal-deletion cases of one reference. Columns compare no alignment, the production range, and the wider search; dark grey, blue (dashed outline), and red (solid outline) denote overlap, missing reference ink, and extra candidate ink. The right-hand column plots the score drop (100 minus score) of each condition on one 0–100 axis, with an arrow from no alignment to the production range; rows (a)–(b) are nuisance perturbations whose drop should vanish, row (c) a structural edit whose drop should remain. Registration removes the translation penalty in row (a), reduces but does not eliminate the scale penalty in row (b), and retains the structural penalty in row (c). Source reference: Calli-Tongji open subset [19], CC BY-NC 4.0.

Table 5 Association between the structural scores and the mean rating of three blinded calligraphy-domain raters across 150 canonical pairs. Confidence intervals use 10,000 bootstrap resamples clustered by character.

Score	ρ	95% CI	p
Production	0.297	[0.148, 0.441]	2.28×10^{-4}
Coverage-aware	0.429	[0.313, 0.535]	4.32×10^{-8}

48.9% exact agreement, 93.3% agreement within one point, MAD 0.578, and pooled QWK 0.634.

ASDS reaches $\rho = 0.556$ (0.454–0.651), and character-grouped out-of-fold weight selection retains $\rho = 0.540$ (0.432–0.639). The paired

intervals for ASDS minus the production and coverage-aware scores are 0.110–0.410 and 0.017–0.239; the out-of-fold difference from the coverage-aware score is less certain (−0.003–0.224). The polar component is the strongest single term ($\rho = 0.547$); grid and projection reach 0.447 and 0.445. Because the same ratings informed feature design, $\rho = 0.556$ is a retrospective development result, and $\rho = 0.540$ is a character-grouped internal-validation estimate that removes weight-selection optimism only: the feature families were also chosen after the ratings, so neither value is an external or prospective confirmation.

The direct-ink control settles what the six-channel masks contribute to this scalar.

Table 6 Calligraphy-domain-rater reliability. ICC uses only the 150 canonical presentations. Repeat statistics use the 15 hidden repeats per rater: Exact and Within ± 1 are agreement fractions between canonical and repeat ratings, MAD the mean absolute difference, and QWK the quadratic weighted kappa; E01–E03 are blinded rater codes and POOLED aggregates all repeats.

Rater	Exact	Within ± 1	MAD	QWK
E01	0.600	1.000	0.400	0.559
E02	0.467	1.000	0.533	0.721
E03	0.400	0.800	0.800	0.591
POOLED	0.489	0.933	0.578	0.634

Canonical ICC(2,1)=0.448 and ICC(2,k)=0.709 for 150 pairs and 3 raters.

Table 7 Development-stage ASDS association with mean human rating (character-cluster bootstrap 95% CI). Grouped OOF: character-grouped out-of-fold weight selection.

Score or component	Spearman ρ	95% CI
Grid only	0.447	[0.303, 0.572]
Projection only	0.445	[0.316, 0.564]
Polar only	0.547	[0.443, 0.645]
ASDS (frozen)	0.556	[0.454, 0.651]
ASDS (grouped OOF)	0.540	[0.432, 0.639]

Table 8 Direct-ink ablation of the frozen ASDS specification on the same 150 development pairs.

ASDS input	Spearman ρ	95% CI
Annotated-mask union	0.556	[0.454, 0.651]
Direct ink	0.556	[0.454, 0.649]
$\Delta\rho$ (annotated–direct)	0.000	[-0.010, 0.010]

Direct ink: grayscale intensity < 240 (frozen QC threshold, Sect. 5.3); same registration, features, and weights for both variants.

Annotated-mask-union and direct-ink ASDS reach $\rho = 0.55623$ and 0.55583 ; their difference is $\Delta\rho = 0.00040$ (95% CI -0.01011 – 0.01030 ; Table 8 rounds to three decimals), and the two score vectors have rank association 0.9973 (Fig. 6). On clean direct-digital input, ASDS is a silhouette score for which thresholded ink and the annotated-mask union are interchangeable.

Answer to RQ3. Blinded ratings of same-character structural similarity associate modestly with the frozen production score ($\rho = 0.297$), more

Table 9 Primary taxonomy of exact-region diagnostic failures. Fractions are shares of the 3,565 exact-region failures.

Failure type	Count	Fraction
Multi-region/boundary truth	1922	0.539
Wrong direction channel	876	0.246
Local finding absent from top 3	718	0.201
Wrong missing/extra type	38	0.011
Alignment residual shift	11	0.003

strongly with the coverage-aware score (0.429), and most strongly with ASDS (0.556 in retrospective development; 0.540 under character-grouped internal validation, where its advantage over the coverage-aware score is not resolved, paired 95% CI -0.003 – 0.224). The ASDS scalar does not depend on six-channel parsing: computed from thresholded ink or from the union of the annotated direction masks, it is equivalent within $\Delta\rho = 0.0004$. The scalar therefore cannot justify the parser. The parser’s role is to supply the direction-level, overlap, endpoint, and localized evidence that a silhouette cannot; its accuracy is measured in RQ1 and the reliability of its localized outputs in RQ4. This is the third contribution. Its boundary is that the supported construct is human-audited structural agreement judged by three domain-qualified raters, two of them authors, on one retrospective cohort; it is not an aesthetic grade.

6.4 RQ4 (localized diagnosis)

RQ4 is answered on the diagnostic cohort with the metrics of Sect. 5.4 (Table 9, Fig. 7).

Required Recall@3 is 0.701, whereas exact grid localization is 0.109. Among exact-region failures, 53.9% arise because the true change spans several cells and the metric demands a single exact cell, 24.6% attribute the finding to the wrong direction channel, and 20.1% omit the local finding from the top three. Figure 7 shows one exact success and one audited failure of the first kind: the returned cell overlaps the multi-cell change, and the frozen metric still records a failure. The dominant failure is thus a mismatch between the localization representation and the evaluation metric, and the second is direction attribution; neither is a wording problem.

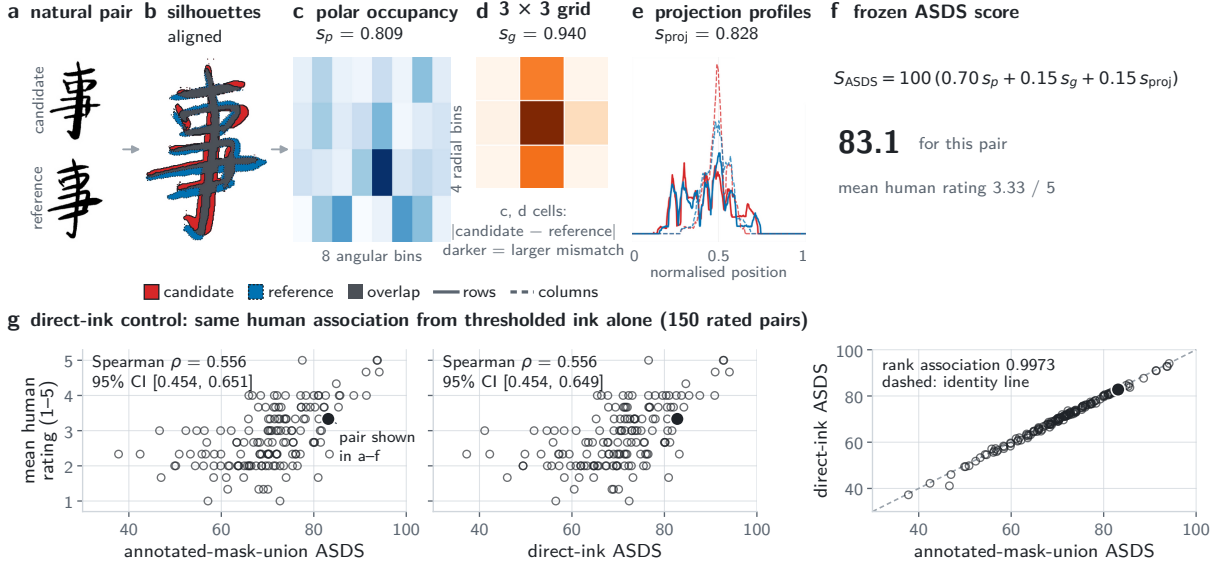


Fig. 6 ASDS mechanism and direct-ink control. Panels (a)–(f) decompose one natural pair into aligned silhouettes, polar and 3×3 occupancy differences (darker cells mark larger $|\text{candidate} - \text{reference}|$), projection profiles (candidate red, reference blue; rows solid, columns dashed), and the frozen formula; panel (g) plots the 150-pair comparison: mean human rating against the annotated-mask-union ASDS (the union of the five annotated direction masks of the project samples, not a parser output) and against the direct-ink ASDS, and the two scores against each other; the filled marker is the pair shown in (a)–(f) and the dashed line the identity. The two inputs have nearly identical human association and score ordering.

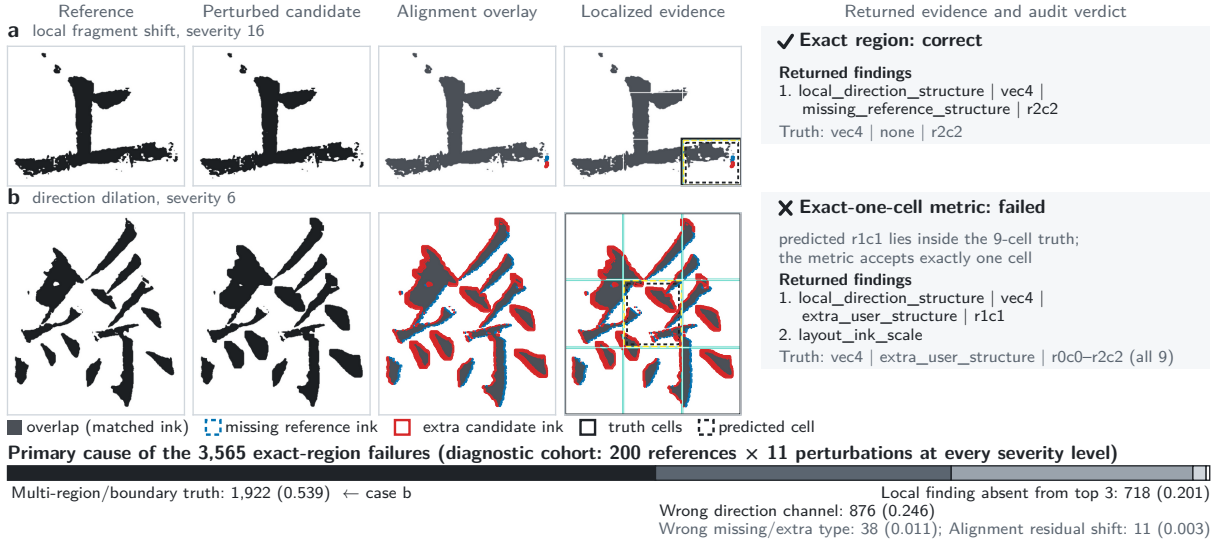


Fig. 7 Frozen diagnostic evidence: one exact-localization success and one audited failure caused by multi-cell truth under an exact-one-cell metric. Findings are printed as finding type | direction channel | missing/extra type | cell, where rXcY indexes the row and column of the 3×3 grid and user denotes the candidate (Sect. 4). Dark grey denotes overlap, blue (dashed outline) missing reference ink, and red (solid outline) extra candidate ink; a solid black frame encloses the truth cells and a dashed black frame the predicted cell. Each card states the audit verdict, the returned findings, and the truth. The bottom strip restates the primary-cause taxonomy of Table 9 over the 3,565 exact-region failures; case (b) belongs to its largest class.

Answer to RQ4. The evidence layer usually surfaces the required finding (Recall@3 of 0.701) but

does not reliably pin it to one 3×3 cell (0.109). Among the exact-cell failures, single-cell scoring

of multi-cell truth and wrong direction attribution dominate; a wrong missing/extra type is the primary cause of only 1.1%, a lower bound on type errors because each failure carries one cause in a fixed order. This is the fourth contribution: a diagnostic failure taxonomy that separates structured evidence from language realization. The direction-level findings evaluated here, together with the endpoint agreement measured in RQ1, are precisely the outputs that the direct-ink control of RQ3 cannot provide. Because the truth is an engineered intervention rather than a teacher’s judgment, the result bounds the mechanism, not pedagogical usefulness.

7 General Discussion and Limitations

7.1 What the four answers imply together

Read together, the answers describe a measurement design rather than a single model. First, low-resource dense parsing must be evaluated on the generalization axis the application needs, since in-corpus fit did not predict transfer to unseen characters, and only overlapping labels can express the crossing evidence of Fig. 4. Second, registration is part of the measurement, not neutral preprocessing: the transform family decides which differences remain visible, and the bounded search was chosen so that global placement, size, and tilt are compensated while local structural loss stays in view.

Third, the scalar and the diagnostics have different justifications. A direction-label swap that preserves the silhouette would leave ASDS unchanged while altering every direction-level, endpoint, and localized output; a system that needs only a scalar silhouette comparison may use thresholded ink, whereas a diagnostic system requires the parser.

The next steps follow: the frozen ASDS specification should be tested without modification on new writers, characters, and raters; and diagnostic localization should be reported with a region-overlap or pixel-overlap metric that credits multi-cell truth.

7.2 Limitations

The segmentation corpus contains 40 labelled character identities from one recovered acquisition source; character-disjoint evaluation is not writer-disjoint evaluation (Sect. 3.1), and collection-specific regularities may survive the semantic quality control.

Controlled perturbations operate on cached reference masks predicted by the v1 checkpoint, which predates the semantic quality control and the formal comparison; any systematic parsing error in those masks propagates into every RQ2 and RQ4 statistic, and an end-to-end failure may combine parsing, registration, and comparison errors. Scale-up and compound summaries are conditional on clipping-safe references because of their high invalid fractions, and the external library supports only seven same-character cross-style pairs.

The human cohort contains 150 pairs rated by three domain-qualified raters, two of whom are authors; single-rater reliability is poor and the three-rater mean moderate; all scores on the cohort are computed from annotated masks, so parser error in deployment is not included; ASDS was developed after the ratings were available, so its association is retrospective and its validation character-grouped rather than external; the contribution of registration to ASDS was not isolated (the polar term is centroid- and scale-normalized, so much of its association may survive without alignment); and the direct-ink control reuses the same cohort. Agreement at multi-label crossing pixels is not measured separately. The diagnostic benchmark uses engineered interventions rather than teacher judgments. Finally, the target modality is clean direct digital writing; camera capture lies outside the claims (Sect. 1).

8 Conclusion

We formulated the assessment of a hand-written Chinese character against a same-character reference as overlapping document parsing, discrepancy-preserving registration, and human-audited comparison, and answered four research questions under limited labelled data. For RQ1 (representation and generalization), six independently activated channels let U-Net,

DeepLabV3+, and SegFormer-B2 recover overlapping direction and endpoint masks from a 769-sample quality-controlled corpus; DeepLabV3+ leads standard-split direction parsing, SegFormer-B2 leads character-disjoint direction parsing, and U-Net leads strict endpoint localization. For RQ2 (registration and score behaviour), constrained similarity registration reduces nuisance sensitivity by 30.118 points relative to no alignment while preserving 11.189 additional points of structural penalty, and a wider search does not improve the trade-off; the claim is relative and conditional on clipping-safe references. For RQ3 (human validity), blinded structural ratings by three domain-qualified raters, two of them authors, associate modestly with the frozen production score, more strongly with the coverage-aware score, and most strongly with ASDS, whose $\rho = 0.556$ is a retrospective development result and whose $\rho = 0.540$ is a character-grouped internal-validation estimate with an unresolved advantage over the coverage-aware score; direct-ink and annotated-mask-union ASDS are equivalent, so the parser’s justification is directional, overlap, endpoint, and localized evidence rather than the scalar. For RQ4 (localized diagnosis), the evidence layer usually surfaces the required finding but does not reliably localize it to one cell; in 53.9% of exact-cell failures the true change spans several cells, which the exact-one-cell metric cannot credit.

The contribution is an auditable framework for reference-conditioned structural assessment in the direct-digital modality: semantic quality control, character-disjoint testing, bounded registration, score-semantics audits, blinded human association with honest labelling of retrospective development, and structured evidence kept upstream of language. It is not a universal aesthetic score.

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Statements and Declarations

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Competing Interests. The authors declare no competing interests.

Ethics Approval. Not applicable. The study analysed images of handwritten characters and perceptual similarity ratings; it involved no clinical, medical, or behavioural intervention on human subjects. The handwriting corpus was commissioned by the project without recording writer identities, and the structural-similarity ratings were provided by two of the authors and one collaborating calligraphy specialist as part of the study design, stored under evaluator codes, with no sensitive personal information collected.

Informed Consent. Not applicable. No identifiable personal data of writers or raters are reported; the three raters took part in the rating study voluntarily as contributors to the project and were informed that the task assessed character structure similarity, not their own ability.

Data Availability. The frozen data contracts and result tables that support the reported values are in the code repository named under Code Availability: the quality-control audit and exclusion contract and both split assignments (`artifacts/data_qc`, `artifacts/data_audit`, `artifacts/paper_ijdar/character_disjoint`); the per-seed segmentation results and checkpoint manifest (`artifacts/paper_ijdar/task1`); the ASDS development features, character-grouped folds, frozen specification, and report (`artifacts/paper_ijdar/spatial_score_development`); the 150-pair direct-ink ASDS table (`artifacts/paper_ijdar/direct_ink_asds`); the reference-library manifest with per-image source paths and SHA-256 digests (`references/calli_tongji_beta_manifest.csv`); and the figure provenance manifest. Online Resource 1, Note S6, lists the digests of every frozen artifact. The external reference images are the Ouyang Xun regular-script and Wang Xizhi running-script categories (100 images each) of the Calli-Tongji open subset [19], publicly downloadable from ModelScope under CC BY-NC 4.0 at <https://www.modelscope.cn/datasets/CalliTo>

ngji/Calli-Tongji_A_Dataset_of_Historical_Calligraphy_Styles; they are reproduced in Figs. 1, 3, and 5 with attribution and are not redistributed. The project-collected handwriting images and six-channel masks, the controlled-perturbation, alignment-ablation, inactive-channel, and cross-reference result files, and the blinded 150-pair rating files (evaluator codes only) are available from the corresponding author on reasonable request for non-commercial research use.

Code Availability. Source code, frozen experiment contracts, analysis scripts, and non-restricted reproducibility artifacts are maintained at <https://github.com/LiuXiaofan-0321/OneStroke2026>; an identity-free archival copy will be supplied if anonymous review is required.

Author Contributions. X.L.: Conceptualization, methodology, software, formal analysis, visualization, writing—original draft, project administration. R.Z.: Data curation, methodology, software, validation, writing—review and editing. Y.F.: Software, system integration, validation, writing—review and editing. All authors contributed to the investigation, approved the submitted manuscript, and accept responsibility for the work.

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